

PAVAN DHARMA ADAPA

Data Engineer · Analytics Engineering

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PROFESSIONAL SUMMARY

Data engineer with an M.S. in Computer Science who builds pipelines other people can run: 34 merged pull requests to an open-source scientific pipeline covering packaging, CI, batch execution and failure isolation, plus SQL and star-schema pipelines over real CMS and CDC data with correctness gates that fail the run rather than ship a wrong number. Work-authorized through June 2027 on F-1 OPT and eligible for the 24-month STEM extension: no sponsorship required until 2029. Available immediately.

TECHNICAL SKILLS

Data Engineering: ETL/ELT, workflow orchestration (Snakemake), batch processing with isolated failures and resumable execution, dimensional modeling and star schemas, SQLite, data quality gates, provenance tracking

Programming & Databases: Python (pandas, NumPy, SciPy), SQL (window functions, CTEs), Bash, R, YAML specs, Advanced Excel

Software & Analysis Practice: Git and GitHub (fork-and-PR under maintainer review), continuous integration (GitHub Actions matrices), pytest, packaging (pip, conda), cross-platform debugging, statistical modeling, Power BI

PROFESSIONAL EXPERIENCE

Data Analytics Intern | *Agricultural Microbiomes Lab, Mississippi State University, Starkville, MS* Jul 2026 – Present

- Primary contributor to **nano16s**, the lab's open-source Nanopore 16S rRNA profiling pipeline (Snakemake, Python, Bash) — **34 merged pull requests, 53 of 64 non-merge commits and +5,700 lines** under maintainer review.
- Built a batch subcommand that processes an entire directory of sequencing runs in one command with per-run logging, isolated failures and resumable execution, removing manual run-by-run invocation for multi-sample projects.
- Fixed two silent data-correctness defects — the wrong abundance table handed downstream, and stray files ingested as samples — the class of bug that corrupts every result built on it without raising an error.

Graduate Teaching Assistant | *Mississippi State University, Starkville, MS* Sep 2025 – May 2026

- Taught data cleaning, transformation and visualization in Python and R to **100+ students across six courses**, and reported completion and grade metrics that let instructors reach at-risk students earlier.

Graduate Research Assistant | *Mississippi State University, Starkville, MS* Aug 2024 – Aug 2025

- Owned a **900,000+ record** public-health surveillance dataset end to end — procurement, validation, cleaning, analysis and reporting — as the single point of accountability for its quality.
- Built the automated collection and cleaning workflow that standardized multi-source research data, **cutting per-dataset processing from hours to under 10 minutes** and making every figure reproducible from source.

DATA ENGINEERING PROJECTS

Self-directed public projects — code, tests and continuous integration on GitHub; the data source is named on each.

Emergency Department Throughput — *CDC/NCHS national survey microdata · Python, SQLite star schema*

- Built a star-schema pipeline over **16,025 real ED visits** survey-weighted to 155,397,747 national visits, with a correctness gate: the run aborts unless it reproduces CDC's published national total, so a weighting error cannot pass.
- Showed between-hospital variation in door-to-provider time is **3.56x larger** than between-hour variation, and reported the widely assumed peak-hour effect as a null result (difference 4.87 min, $p = 0.406$) rather than dropping it.

Claims Denials & Appeals — *CMS Transparency in Coverage 2021–2023 · SQL, window functions, two-grain fact design*

- Modeled **15,545 plan-years** and **126.9M denied claims** across 624 issuer-years in a single fact design carrying two grains, using window functions for year-over-year comparison on real public issuer filings.
- Found only **0.168% of denials are ever appealed while 40.1% of appeals are overturned**, a rate that holds across all three years — locating the bottleneck in filing capacity rather than claim merit.

EDUCATION

M.S., Computer Science — Mississippi State University, Mississippi State, MS May 2026

B.Tech, Computer Science — Malla Reddy University, Hyderabad, India May 2024

Publication (under revision): Li Zhang, Saikanth Ratnavale, Pavan Dharma Adapa, Sai Harika Gade, Michael E. Navicky — Computational Tools for Modeling and Predicting Respiratory Disease Spread in Commercial Poultry Production.